

1. 8-QUEEN PROBLEM

Aim

To write a Python program to solve the 8-Queen problem using Backtracking.

Objective

To place 8 queens on an 8×8 chessboard so that no two queens attack each other.

Algorithm

1. Create an empty 8×8 chessboard.
2. Start from the first column.
3. Place a queen in a safe row.
4. Check rows and diagonals for safety.
5. If safe, move to the next column.
6. If no safe position exists, backtrack.
7. Print the solution.

Python code:

```
N = 8
```

```
board = [["."] * N for _ in range(N)]
```

```
def safe(r, c):
```

```
    for i in range(c):
```

```
        if board[r][i] == "Q":
```

```
            return False
```

```
    i, j = r, c
```

```
    while i >= 0 and j >= 0:
```

```
        if board[i][j] == "Q":
```

```
            return False
```

```
    i -= 1
```

```
    j -= 1
```

```
    i, j = r, c
```

```

while i < N and j >= 0:
    if board[i][j] == "Q":
        return False
    i += 1
    j -= 1
return True

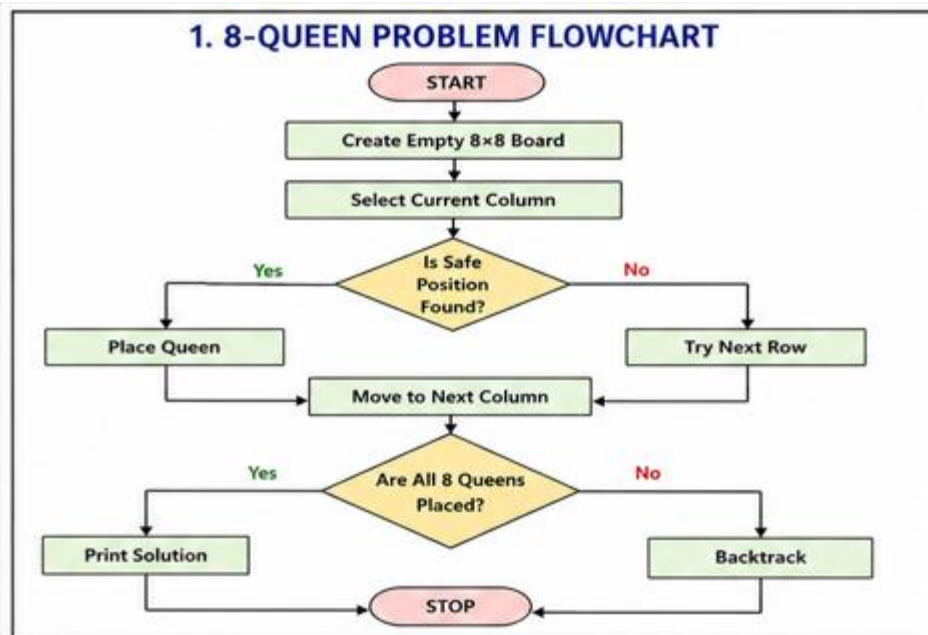
def solve(c):
    if c == N:
        return True
    for r in range(N):
        if safe(r, c):
            board[r][c] = "Q"
            if solve(c + 1):
                return True
            board[r][c] = "."
    return False

solve(0)

for row in board:
    print(*row)

```

Flowchart



Output

```

IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Sheik Mohamed/Downloads/water program.py =====
Q . . . . . Q .
. . . . Q . .
. Q . . . . .
. . . Q . . .
. . . . Q . .
. . Q . . . .
>>>
  
```

Result

Thus, the Python program to solve the 8-Queen problem was executed successfully.

Conclusion

The 8-Queen problem successfully solved

